PROBLEM FRAMING MEMO

Customer Churn Prediction Using Machine Learning

Project: Customer Churn Prediction
Domain: Artificial Intelligence and Machine Learning

1. Introduction

Customer churn refers to a situation where customers stop using a company's products or services. It is a major challenge for businesses because losing customers can reduce revenue and affect business growth.

Machine Learning can help businesses analyze customer data and identify customers who are likely to leave. This project aims to build a basic model that predicts customer churn using historical customer information.

2. Problem Statement

Businesses often find it difficult to identify customers who may stop using their services. Without early identification, companies may lose valuable customers.

The problem is to develop a Machine Learning model that analyzes customer-related data and predicts whether a customer is likely to churn.

3. Project Objectives

- Understand the factors that may influence customer churn.
- Explore and analyze the available customer dataset.
- Prepare the data for Machine Learning.
- Build a baseline classification model to predict churn.
- Evaluate the model using suitable performance metrics.
- Help businesses identify customers who may need additional support or retention offers.

4. Proposed Solution

The project will use a customer churn dataset containing relevant customer details and a churn label.

The data will be cleaned and prepared before training a baseline classification model. The model will learn patterns from existing customer records and predict whether a customer is likely to leave.

The model's performance will be evaluated using accuracy, precision, recall, and F1-score. These metrics will help assess how well the model identifies customers who may churn.

5. Expected Outcome

The expected outcome is a working baseline Machine Learning model that estimates customer churn risk. The results may help businesses plan customer-retention strategies and improve customer satisfaction.

The model will support decision-making, but its predictions should not be treated as guaranteed outcomes.

6. Responsible AI Considerations

- **Privacy:** Customer data should be handled securely and used only for the intended purpose.
- **Fairness:** The model should be checked for unfair differences in prediction performance across customer groups.
- **Transparency:** The model's limitations and evaluation results should be documented.
- **Human oversight:** Business decisions should not rely only on automated predictions.
- **Data quality:** Missing, incorrect, or biased data may affect prediction quality.

7. Success Criteria

The project will be considered successful if it:

- Builds a functioning baseline churn prediction model.
- Evaluates the model using appropriate metrics.
- Documents the dataset, assumptions, and limitations.
- Identifies potential risks associated with using the model.

8. Conclusion

This project demonstrates how Machine Learning can be applied to customer churn prediction. By identifying possible churn patterns, businesses may be able to take early action to improve customer retention. Responsible data handling, fair evaluation, and human oversight will be important throughout the project.